

JUNCHENG LU

jl7191@tc.columbia.edu ♦ 773-328-0805

EDUCATION

Teachers College, Columbia University
M.A. in Cognitive Science in Education
GPA: 4.0/4.0

September 2025 - Present

Beijing Normal University
B.S. in Psychology
GPA: 3.45/4.0

September 2020 - July 2024

Coursework: *Computational Cognitive Neuroscience*
Brain Imaging Data Modeling
fMRI: Principles, Experiment Design and Data Analysis

RESEARCH EXPERIENCE

Research Assistant
Social Cognitive and Neural Sciences Lab
PI: Dr. Jon Freeman

September 2025 - Present
New York, NY

- Conduct computational modeling and model-based analysis of social decision-making and implicit bias tasks, using DDM/MPT approaches and fMRI-informed hypotheses.

Research Assistant
Intergroup Relation Neuroscience Lab
PI: Dr. Xiaochun Han

March 2022 - June 2025
Beijing, China

- Contributed to behavioral, EEG, and computational analyses of intergroup decision-making, leading to conference presentation and manuscript preparation.

PUBLICATIONS

Lu, J.^{*}, Zhang, Y.^{*}, Han, X., Xiao, Z., Han, X.[†] Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. *Revise-and-resubmit at Journal of Experimental Social Psychology.*

Zhang, Y.^{*}, Lu, J.^{*}, Han, X.[†] Cognitive mechanism of killing decision discrimination between combatants and civilians during intergroup conflict: The influence of social role information. *Manuscript in preparation.*

^{*} These authors contributed equally to this work. [†] Corresponding author.

CONFERENCES

Lu, J., Han, X.[†] (2024, April). Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. *Paper accepted for **Oral Presentation** and **Poster Presentation** at the **Social and Affective Neuroscience Society (SANS) 2024**. (Unable to attend due to visa restrictions)*

SELECTED PROJECTS

Awareness of Implicit Bias and Its Cognitive Basis
Independent project under the supervision of a postdoctoral researcher, Jon Freeman Lab

Jan 2026 - Present

- Applied RT-MPT models to multiple Implicit Association Tests (IATs) to decompose performance into latent cognitive processes underlying implicit bias and controlled responding.
- Examined whether individuals' awareness of their own bias was associated with specific MPT parameters, testing which cognitive components contribute to metacognitive insight into biased behavior.
- Conducted model specification, parameter estimation, and model comparison in R, linking self-report measures of bias awareness to latent cognitive parameters.

Computational Modeling of Weapon Identification Task

Sept 2025 - Dec 2025

Independent project under the supervision of a postdoctoral researcher, Jon Freeman Lab

- Applied Drift Diffusion Model (DDM) and Multinomial Processing Tree (MPT) models to behavioral data from the Weapon Identification Task to decompose decision bias into latent cognitive parameters.
- Conducted model specification, parameter estimation, and model comparison across competing cognitive architectures.
- Integrated DDM/MPT parameters with fMRI activation measures to test model-based accounts of threat and control processes in biased decision-making.

Computational and Neural Mechanisms of Punishment Decision-making

Sept 2022 - June 2025

Independent undergraduate thesis, Xiaochun Han Lab

- Applied Drift Diffusion Model (DDM) to decompose intergroup harm-related decisions into latent cognitive components.
- Analyzed EEG data to identify temporal neural markers associated with distinct stages of the decision process.
- Integrated DDM parameters with ERP components to test correspondence between latent decision processes and neural dynamics.
- Contributed to experiment design, data collection, data analysis, model interpretation, and manuscript writing; manuscript currently under review at JESP.

TECHNICAL SKILLS

Programming Languages

Python (HDDM, NumPy, Pandas), R (brms, lme4, TreeBUGS, rtmpt),
MATLAB (EEGLAB, SPM12, Psychtoolbox), HTML/JavaScript (JsPsych)

Computational Methods

Hierarchical Bayesian modeling, Drift Diffusion Model (DDM),
Multinomial Processing Tree (MPT), mixed-effects models

Neuroimaging

fMRI data analysis (SPM12), EEG data collection and analysis (EEGLAB),
brain-behavior correlation analysis